American Computer Science League

2022 Finals ● Short Solutions ● Senior Division

1. Boolean Algebra

$A(\overline{A} + B)\,\overline{C} + \overline{A\,C} + \overline{B + C}$
$= A\,\overline{A}\,\overline{C} + A\,B\,\overline{C} + \overline{A} + \overline{C} + \overline{B}\,\overline{C}$
$= 0 + \overline{C}(A\,B + 1 + \overline{B}) + \overline{A}$
$= \overline{A} + \overline{C}$

B. $\overline{A} + \overline{C}$

2. Boolean Algebra

How many ordered triples make the following expression FALSE?

$\overline{A}\,(\overline{A + B})\,(\overline{A\,B})\,(\overline{A}\,B)\,\overline{C} = \overline{A}\,\overline{A}\,\overline{B}\,(\overline{A} + \overline{B})\,\overline{A}\,B\,\overline{C} = 0$
Therefore 8 ordered triples make it FALSE.

D. 8

3. Bit-String Flicking

(NOT (LCIRC-1 10110) OR (RSHIFT-2 01100))
AND (LSHIFT-1 (RCIRC-2 01101))
= ((NOT 01101 OR 00011) AND (LSHIFT-1 01011))
= ((10010 OR 00011) AND 10110)
= 10011 AND 10110
= 10010

A. 10010

4. Bit-String Flicking

Let X = abcde
LHS = ((LCIRC-3 (10100 OR X)) AND (11010 OR (RSHIFT-1 X)))
= ((LCIRC-3 (10100 OR abcde)) AND
(11010 OR (RSHIFT-1 abcde)))
= ((LCIRC-3 1b1de) AND (11010 OR 0abcd))
= de1b1 AND 11b1d
= debbd
LHS = RHS ⇒ debbd = 11001
So d = 1, e = 1, b = 0, a = *, c = *
Therefore *0*11 makes it TRUE which is 4.

C. 4

5. Recursive Functions

Find $f(f(f(f(20))))$ given:

$f(20) = f(10) + 3 = 7 + 3 = 10$ $\quad f(f(f(f(20)))) = f(f(f(10)))$
$f(10) = f(5) + 3 = 4 + 3 = 7$ $\quad = f(f(f(7)))$
$f(5) = f(3) + 1 = 3 + 1 = 4$ $\quad = f(f(5))$
$f(3) = f(1) + 1 = 2 + 1 = 3$ $\quad = f(5)$
$f(1) = 1 + 1 = 2$ $\quad = 4$
$f(7) = f(5) + 1 = 4 + 1 = 5$

A. 4

6. Recursive Functions

For mod 4 the sequence is:

0, 1, 1, 2, 3, 1, 0, 1, 1, 2, 3, 1, …

Therefore, the Pisano period for mod 4 is 6.

B. 6

7. Digital Electronics

The circuit is: $\overline{A + B}$ $gate?$ $\overline{\overline{B\,C} + C}$
$= \overline{A}\,\overline{B}$ $gate?$ $B\,C\,\overline{C}$
$= \overline{A}\,\overline{B}$ $gate?$ 0

Gates: AND - all ordered triples make it FALSE
OR - 6 make it FALSE
NAND - 0 make it FALSE
NOR - 2 make it FALSE

A. AND

8. Digital Electronics

The circuit translates to: $\overline{A + (A \oplus B)(B + \overline{\overline{B\,C}})}$

$\overline{A + (A \oplus B)(B + \overline{\overline{B\,C}})}$
$= \overline{A + (A \oplus B)(B + \overline{B} + \overline{C})}$
$= \overline{A + (A \oplus B)}$
$= \overline{A}\,\overline{(A \oplus B)}$
$= \overline{A}\,(\overline{A} \oplus B)$
$= \overline{A}\,(A\,B + \overline{A}\,\overline{B})$
$= \overline{A}\,A\,B + \overline{A}\,\overline{B}$
$= \overline{A}\,\overline{B}$
$= \overline{A + B}$

D. $\overline{A + B}$

9. Prefix-Infix-Postfix

+ + / * # 1 2 3 4 6 * $ 8 4 2 9 # 3 $ 4 5 6 2
= + + / * (# 1 2 3) 4 6 * ($ 8 4 2) 9 # 3 ($ 4 5 6) 2
= + + / (* 3 4) 6 (* 2 9) (# 3 4 2)
= + + (/ 12 6) 18 4
= + (+ 2 18) 4
= + 20 4
= 24

C. 24

10. Prefix-Infix-Postfix

$$\cfrac{1}{x + \cfrac{1}{x + \cfrac{1}{x+1}}}$$

= 1 / (x + (1 / (x + (1 / (x + 1)))))
= 1 / (x + (1 / (x + (1 / (x 1 +)))))
= 1 / (x + (1 / (x + (1 x 1 + /))))
= 1 / (x + (1 / (x 1 x 1 + / +)))
= 1 / (x + (1 x 1 x 1 + / + /))
= 1 / (x 1 x 1 x 1 + / + / +)
= 1 x 1 x 1 x 1 + / + / + /

A. 1 x 1 x 1 x 1 + / + / + /

11. Computer Number Systems

$2022_{10} = 7E6_{16}$
The next 3 consecutive hexadecimal digits would be:
$89A_{16} = 2202$ (increasing) or $876_{16} = 2166$ (decreasing)
Therefore the next year after 2022 is 2166.

D. 2166

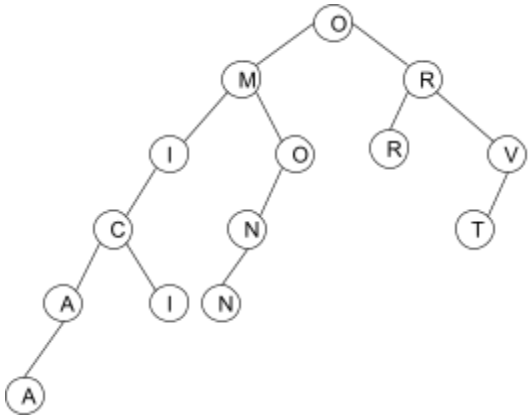
12. Computer Number Systems

$50_{10} = 62_8$ and $100_{10} = 144_8$
The octal numbers containing a 2 are: 62, 72, 102, 112, 120, 121, 122, 123 124, 125, 126, 127, 132, and 142.
Therefore there are 15 2's.

C. 15

13. Data Structures

The binary search tree for **OMICRONVARIANT is:**



The internal path length is: 2 * 1 + 4 * 2 + 3 * 3 + 3 * 4 + 1 * 5 = 2 + 8 + 9 + 12 + 5 = 36

B. 36

14. Data Structures

The STACK is constructed using LIFO as follows:
A, EMPTY, D, DV, DVE, DV, D, DN, DNT, DNTU, DNTUR, DNTU, DNT, DNTO, DNTOU, DNTOUS, DNTOU, DNTO, DNT, DN, D
The next item popped is D.

The QUEUE is constructed using FIFO s follows:
A, EMPTY, D, DV, DVE, VE, E, EN, ENT, ENTU, ENTUR, NTUR, TUR, TURO, TUROU, TUROUS, UROUS, ROUS, OUS, US, S
The next item popped is S.

S is alphabetically after D.

B. S

15. Graph Theory

$$\begin{vmatrix} 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \end{vmatrix}^{2} = \begin{vmatrix} 1 & 2 & 1 & 1 & 1 \\ 2 & 3 & 0 & 0 & 0 \\ 0 & 1 & 2 & 2 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 2 & 2 & 1 \end{vmatrix}$$

There are 26 paths of length 2.

C. 26

16. Graph Theory There are 4 by inspection: ABCDA, ABEDA, ACBEA, and ACDEA.	C. 4
17. What Does This Program Do? There are 5 Fibonacci numbers which are divisible by 2, 3 that are divisible by 3, 2 that are divisible by 5, and 2 that are divisible by 8 for the numbers given. 5+3+2+2 = 12.	A. 12
18. LISP (CADADAR W) = (CADADAR (REVERSE Z)) = (CADADAR (REVERSE (CONS Y X))) = (CADADAR (REVERSE (CONS Y '((1 2 (3)) (2) 3 (1(2 (3))))))) = (CADADAR (REVERSE '((2 1 (3 (3))) (1 2 (3)) (2) 3 (1 (2 (3)))))) = (CADADAR '((1 (2 (3))) 3 (2) (1 2 (3)) (2 1 (3 (3))))) = (CADADR '(1 (2 (3)))) = (CADAR '((2 (3)))) = (CADR '(2 (3))) = (CAR '((3))) = (3)	D. (3)
19. FSAs and Regular Expressions **[^iop] [aeiou] [a-g]* [m-t]* (e|s)*** Only 3 items satisfy the regular expression: seeds weeder, hoses. mulch - fails at l rake - fails at k daylily - fails at y ivy - fails at i pots - fails at p berries - fails at i mower - fails at w grass - fails at r straw - fails at t	B. 3
20. Assembly Language This program finds the sum of the squares of each Fibonacci number in the sequence 8, 5, 3, 2, 1. It is always true that this sum is 1 less than the product of the first number and the next one, 13. In general, the product minus the total is always 1. (8*13) - (64+25+9+4+1) = 1.	B. 1